

Efficiency vs. Accuracy in Emotion Recognition

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Abstract

Large language models can often learn new tasks from just a few examples (Few-Shot), without any extra training, but they are very expensive to run in practice. Smaller models like DistilBERT are much cheaper and faster, yet they usually need a lot of labeled data before they perform well. This project focuses on a fine-grained emotion classification task and asks a simple question: at what point does a fine-tuned small model actually become better than a large few-shot model? We call this the *Breakeven Point*.

1 Problem Statement

Large language models can often learn new tasks from just a few examples (Few-Shot), without any extra training, but they are very expensive to run in practice. Smaller models like DistilBERT are much cheaper and faster, yet they usually need a lot of labeled data before they perform well. This creates a practical trade-off: either rely on an expensive, general-purpose model with minimal setup, or invest in collecting data and fine-tuning a smaller model.

In this project, we focus on a fine-grained emotion classification task and ask a simple question: at what point does a fine-tuned small model actually become better than a large few-shot model? We refer to this as the **Breakeven Point**, defined as the number of labeled training examples where the fine-tuned model first surpasses the few-shot baseline.

2 Dataset

To conduct this experiment, the project will utilize the GoEmotions dataset (Demszky et al., 2020), a corpus of Reddit comments originally labeled with 27 emotion categories. To ensure a manageable experimental setup and in the interest of time, we won't use all 27 classes but cut it down to around 10–15.

3 Initial Experiments

We set up two classification pipelines.

Pipeline 1. The first pipeline represents the few-shot prompting approach (Brown et al., 2020), utilizing a large baseline model such as Mistral-7B or Llama-3-8B. Through in-context learning, this model will be provided with a prompt containing X randomly selected examples and asked to classify target sentences without undergoing any actual training or weight updates.

Pipeline 2. The second pipeline represents the fine-tuning approach, utilizing the significantly smaller DistilBERT-base-uncased model (Sanh et al., 2019). Independent versions of this model will undergo full fine-tuning on incrementally larger subsets of the training data: $n = 50$, $n = 100$, $n = 500$, $n = 1000$, and the complete dataset.

Both pipelines will then be evaluated on a held-out testing set using F1-Score and Accuracy metrics.

The primary deliverable of this research will be a **Data Efficiency Curve** plotting the number of training examples (n) against model performance (accuracy). This visualization will help identify the exact Breakeven Point where the fine-tuned model surpasses the prompted baseline, and this will conclude our project.

Limitations

This project is scoped to a reduced subset of the GoEmotions label space and a fixed set of training size checkpoints ($n = 50, 100, 500, 1000$), so results may not generalize to all emotion taxonomies or data regimes.

References

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- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. [DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter](#). *NeurIPS Workshops*.